



SYSTEM PROPOSAL

DMS System for East Shoa Investment and Industry

Manage government office data in best ways

**SUBMITTED TO: East shewa zone Investment and
Industry manager**

ICT DEPARTMENT

PROPOSAL TYPE: Government System Design and Development

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LOCATION: Adama, Ethiopia

Declaration

We hereby declare that the **DMS System** titled in this report is an original work and has not been submitted, published, or presented by any individual or organization. All the content, design, and implementation presented in this project are the result of the collective effort of the members listed below.

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CHAPTER ONE

INTRODUCTION

1.1 Introduction

The **Document Management System (DMS)** is a secure, web-based information system designed to manage, store, organize, track, and control digital documents within an organization. This system is developed specifically for the **East Shoa Zone Investment and Industry** to modernize document handling processes and improve efficiency, security, and accessibility of organizational records.

In many government institutions, including the East Shoa Zone Investment and Industry manager, document management is still largely dependent on manual, paper-based processes. These traditional methods are characterized by excessive paperwork, poor document tracking, limited access control, and a high risk of document loss or damage. As document volume grows, managing files efficiently becomes increasingly difficult, leading to operational inefficiencies and delays in decision-making.

To address these challenges, the proposed DMS introduces a centralized digital platform that enables authorized users to create, organize, access, share, and manage documents based on predefined roles and permissions. The system employs **Role-Based Access Control (RBAC)** to ensure that users can only perform actions aligned with their responsibilities. By leveraging modern web technologies such as **Next.js, TypeScript, and MySQL**, the system provides a scalable, secure, and user-friendly solution for effective document management.

Ultimately, the Document Management System aims to enhance productivity, improve data security, and support transparent and accountable document handling practices within the East Shoa Zone Investment and Industry manager.

1.1.1 Background of the Organization

The **East Shoa Zone Investment and Industry manager** is a government institution responsible for coordinating administrative activities, managing public records, and overseeing zonal-level operations in accordance with regional and national policies. The Investment and Industry manager handles a large volume of official documents, including reports, correspondence, policies, and operational records, which are essential for governance and decision-making.

With a strong commitment to transparency, accountability, and efficient public Investment and Industry manager, the East Shoa Zone Investment and Industry manager continuously seeks to improve its operational processes. However, the reliance on traditional paper-based document management has created challenges in document storage, retrieval, security, and long-term preservation.

Recognizing the growing importance of digital transformation, the East Shoa Zone Investment and Industry manager initiated the development of a Document Management System as a strategic step toward modernizing internal document handling processes. By adopting modern information technologies and structured access control mechanisms, the organization aims to improve document availability, reduce operational risks, and enhance overall administrative efficiency.

1.1.2 Existing System Overview

The existing document management process within the East Shoa Zone Investment and Industry manager is predominantly manual. Documents are created, stored, and managed in physical files, cabinets, and folders, with limited use of digital tools. This approach has resulted in several operational challenges.

Document retrieval is time-consuming and often requires manual searching through physical files. Tracking document history, ownership, and status is difficult, leading to duplication of documents and inconsistencies across departments. Additionally, sharing documents between offices relies on physical transfers or informal communication channels, increasing the risk of document loss and unauthorized access.

From a security perspective, the existing system provides minimal protection against unauthorized access, document tampering, or accidental deletion. Sensitive documents are vulnerable to

damage, loss, or misuse. The lack of proper access control mechanisms further complicates accountability and auditability.

These limitations highlight the need for a centralized, secure, and role-based digital document management solution.

1.1.3 Statement of the Problem

The East Shoa Zone Investment and Industry manager faces significant challenges due to inefficient and outdated document management practices. The continued reliance on manual processes has negatively impacted productivity, security, and organizational effectiveness.

1.1.3.1 Challenges for the Investment and Industry manager

- ❖ **Inefficient Document Handling:** Manual creation, storage, and retrieval of documents consume excessive time and effort.
- ❖ **Poor Record Organization:** Lack of structured storage leads to misplaced or duplicated documents.
- ❖ **Limited Access Control:** Absence of role-based permissions allows unauthorized access to sensitive documents.
- ❖ **Security Risks:** Physical documents are exposed to loss, damage, and unauthorized modification.
- ❖ **Lack of Accountability:** Difficulty in tracking document ownership and modification history.

1.1.3.2 Challenges for Staff

- ❖ **Delayed Access to Documents:** Staff members experience delays when retrieving required files.
- ❖ **Ineffective Collaboration:** Sharing documents across departments is slow and unreliable.
- ❖ **Increased Workload:** Manual document handling increases administrative burden.

1.1.3.3 Overall Systemic Issues

- ❖ Fragmented document storage systems
- ❖ Lack of audit trails and version control
- ❖ Inefficient use of digital technology
- ❖ Reduced organizational productivity and transparency

These challenges demonstrate the urgent need for a modern Document Management System tailored to the operational requirements of the East Shoa Zone Investment and Industry manager .

1.1.4 Proposed System

The proposed **Document Management System (DMS)** is a centralized, web-based solution designed to digitize and streamline document management processes within the East Shoa Zone Investment and Industry manager . The system provides structured document storage, secure access control, and efficient collaboration through role-based permissions.

A key feature of the proposed system is the implementation of **Role-Based Access Control (RBAC)**, allowing administrators and managers to define roles, assign permissions, and control user access dynamically. The system supports multiple user roles, including **Manager, Officer, and Store Keeper**, each with specific access rights.

1.1.4.1 Key Features and Functionalities

❖ **User, Role, and Permission Management (RBAC)**

Managers can create custom roles, assign permissions, and manage users based on organizational requirements.

❖ **Folder Management**

Authorized users can create, update, view, and delete folders according to their assigned permissions.

❖ **File Management**

The system supports file upload, viewing, editing, sharing, and deletion with soft-delete and permanent delete mechanisms.

❖ **Trash Management**

Soft-deleted files are moved to a Trash area where managers can restore or permanently delete documents.

❖ **File Sharing**

Managers can securely share files with selected users based on defined permissions.

❖ **Audit and Access Control**

The system tracks document activities to ensure accountability and traceability.

1.1.4.2 Advantages of the Proposed System

- ❖ Improved document organization and retrieval
- ❖ Enhanced data security through RBAC
- ❖ Reduced risk of document loss
- ❖ Increased operational efficiency
- ❖ Better accountability and transparency

1.2 Objectives

1.2.1 General Objective

The general objective of this project is to design and develop a secure, efficient, and role-based Document Management System for the East Shoa Zone Investment and Industry manager to improve document handling, accessibility, and security.

1.2.2 Specific Objectives

For Managers

- ❖ Manage users, roles, and permissions
- ❖ Control document access and sharing
- ❖ Restore or permanently delete documents

For Officers

- ❖ Create and manage folders and files
- ❖ Perform soft deletion of documents

For Store Keepers

- ❖ View and access documents in read-only mode

1.3 Scope and Limitations

1.3.1 Scope

- ❖ User, role, and permission management
- ❖ Folder and file management
- ❖ Document sharing and access control
- ❖ Trash and recovery mechanism
- ❖ Secure authentication and authorization

1.3.2 Limitations

- ❖ Initial deployment limited to East Shoa Zone Investment and Industry manager
- ❖ Internet dependency
- ❖ No integration with external document systems in the initial phase

1.4 Methodology

- ❖ Requirement analysis through interviews and document review
- ❖ System modeling using UML diagrams
- ❖ Agile-based development approach
- ❖ User-centered design principles

1.5 Feasibility Study

- ❖ **Technical Feasibility:** Uses modern, widely supported web technologies
- ❖ **Operational Feasibility:** Aligns with administrative workflows
- ❖ **Legal Feasibility:** Ensures data protection and access control
- ❖ **Political Feasibility:** Supports digital governance initiatives

CHAPTER TWO

SYSTEM REQUIREMENTS SPECIFICATION (SRS)

2.1 Background (Overview)

The System Requirements Specification (SRS) defines the functional and non-functional requirements for the **Document Management System (DMS)** developed for the **East Shoa Zone Investment and Industry manager**. This document serves as a formal reference for system developers, project supervisors, and stakeholders by clearly describing system behavior, constraints, interfaces, and user interactions.

The DMS is designed to replace manual and semi-digital document handling practices with a centralized, secure, and role-based digital platform. The system enables authorized users to create, organize, access, share, and manage documents efficiently while enforcing strict access control through **Role-Based Access Control (RBAC)**.

The platform is implemented using **Next.js** for the frontend and **Node.js with Express** for the backend, with **JWT-based authentication**, **Prisma ORM**, and **MySQL** as the database. The system supports multiple user roles, including **Manager, Officer, and Store Keeper**, each with defined permissions.

2.2 Scope and Purpose

2.2.1 Scope

The scope of the Document Management System for the East Shoa Zone Investment and Industry manager includes the development of a web-based platform that supports secure and efficient document handling. The system provides functionality for:

- Centralized document and folder management
- Role-based access control (RBAC)
- User, role, and permission management
- File sharing and controlled access

- Soft delete (Trash) and permanent delete mechanisms
- Secure authentication and authorization

2.2.2 Purpose

The purpose of this project is to design and implement a secure and efficient Document Management System that improves document organization, accessibility, and security within the East Shoa Zone Investment and Industry manager . By digitizing document workflows and enforcing access control policies, the system aims to reduce document loss, improve productivity, and ensure accountability in administrative operations.

2.3 Overall Description of Software Requirements

2.3.1 Product Perspective

The Document Management System functions as a centralized document repository for the East Shoa Zone Investment and Industry manager . It replaces fragmented and paper-based storage with a unified digital platform.

Key perspectives include:

- **Integration:** Integrates authentication, authorization, and document storage within a single platform
- **Scalability:** Supports increasing numbers of users, documents, and departments
- **Customization:** Allows managers to define roles and permissions dynamically
- **Security:** Ensures data confidentiality through JWT authentication and RBAC

2.3.2 User Characteristics

Manager

- Has advanced system privileges
- Can manage users, roles, and permissions
- Possesses intermediate to advanced technical skills

Officer

- Responsible for creating and managing folders and files
- Has basic to intermediate technical skills

Store Keeper

- Has read-only access to documents
- Requires simple and intuitive interfaces

2.4 General Constraints

2.4.1 Software Constraints

- Browser compatibility: Chrome, Firefox, Edge
- Frontend: Next.js with Tailwind CSS
- Backend: Node.js with Express
- Authentication: JWT
- Database: MySQL with Prisma ORM

2.4.2 Hardware Constraints

- Access via desktop, laptop, tablet, or smartphone
- Server hosted on VPS infrastructure
- Reliable internet connection required

2.4.3 Assumptions and Dependencies

Assumptions

- Users have basic computer and internet access
- Users are assigned appropriate roles
- Organization supports digital document workflows

Dependencies

- Dependence on server availability

- Dependence on database integrity and backups

2.5 Specific Requirements

2.5.1 User Requirements

Users of the Document Management System must be able to:

- Authenticate securely using username and password
- Access features based on assigned roles and permissions
- Upload, view, and manage documents according to access rights
- Organize documents into folders
- Restore or permanently delete documents based on permission level

2.5.2 Functional Requirements

Manager

- **FR1:** Manager can register, update, and delete users
- **FR2:** Manager can create roles and assign permissions (RBAC)
- **FR3:** Manager can create, edit, and delete folders
- **FR4:** Manager can upload, edit, share, and delete files
- **FR5:** Manager can restore or permanently delete files from Trash

Officer

- **FR6:** Officer can create, edit, and delete folders
- **FR7:** Officer can upload, edit, and view files
- **FR8:** Officer can soft-delete files (move to Trash)
- **FR9:** Officer cannot permanently delete files

Store Keeper

- **FR10:** Store Keeper can view and download files

- **FR11:** Store Keeper cannot create, edit, or delete files or folders

Authentication and Authorization

- **FR12:** System must authenticate users using JWT
- **FR13:** System must enforce role-based permissions on all actions

2.5.3 Non-Functional Requirements

- **Performance:** Handle concurrent users without performance degradation
- **Security:** JWT authentication, RBAC, encrypted communication (HTTPS)
- **Scalability:** Support future expansion of users and documents
- **Availability:** System accessible during working hours
- **Usability:** Simple and intuitive UI
- **Maintainability:** Modular backend and frontend architecture

2.6 External Interface Requirements

2.6.1 User Interfaces

Manager Interface

- User, role, and permission management dashboard
- Folder and file management panel
- Trash management

Officer Interface

- Folder and file management dashboard
- Soft delete functionality

Store Keeper Interface

- Read-only document viewing interface

2.6.2 Hardware Interfaces

- VPS Server
- Minimum 8 GB RAM with 4 CPU
- SSD storage (500 GB or more)
- Keyboard, mouse, and internet connectivity

2.6.3 Software Interfaces

- **Frontend:** Next.js with Tailwind CSS
- **Backend:** Node.js + Express
- **Database:** MySQL
- **ORM:** Prisma
- **Authentication:** JWT

2.6.4 Communication Interfaces

- HTTPS for secure data transmission
- RESTful APIs for frontend-backend communication

CHAPTER THREE

REQUIREMENT ANALYSIS AND MODELING

3.1 Overview of the Analysis Model

The requirement analysis and modeling phase provides a structured understanding of the **Document Management System (DMS)** by translating user needs and system requirements into visual and logical models. This phase ensures that system functionality, workflows, and interactions are clearly defined before implementation.

The analysis model helps to:

- Identify system actors and their responsibilities
- Define how users interact with the system
- Visualize document workflows and access control mechanisms
- Ensure alignment with functional and non-functional requirements

The analysis includes the following modeling techniques:

- **Use Case Diagrams:** Illustrate interactions between users and the DMS
- **Activity Diagrams:** Describe workflows such as login, file upload, deletion, and restoration
- **Entity Relationship Diagram (ERD):** Defines database structure and relationships
- **Sequence Diagrams:** Show step-by-step interaction between system components

These models serve as a reference for developers, system analysts, and stakeholders.

3.2 System Use Case Diagram

The Use Case Diagram illustrates how different users interact with the **Document Management System** based on their roles and permissions.

Actors

1. **Manager**
 - Has full control over the system
 - Manages users, roles, and permissions
 - Manages folders and files
 - Restores and permanently deletes files
2. **Officer**
 - Creates and manages folders

- Uploads and manages files
- Performs soft deletion of files

3. Store Keeper

- Views and downloads documents only
- Has read-only access

Use Cases

Manager

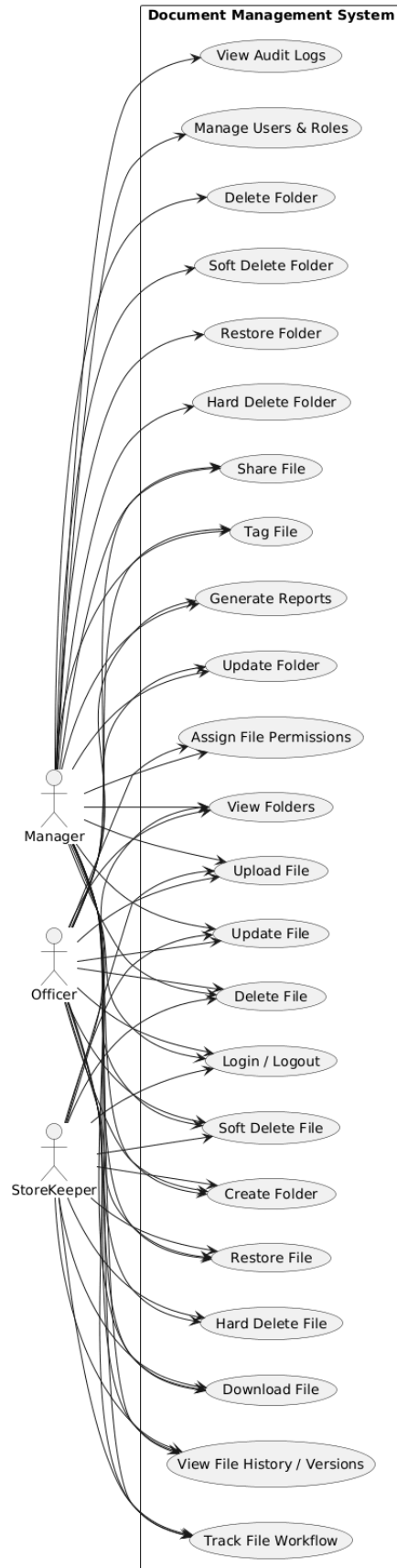
- Create, update, and delete users
- Create roles and assign permissions (RBAC)
- Create, update, and delete folders
- Upload, edit, delete, and share files
- Restore files from Trash
- Permanently delete files

Officer

- Create and manage folders
- Upload and edit files
- Soft delete files (move to Trash)
- View files

Store Keeper

- View files
- Download files



✦ **Figure 1:** Use Case Diagram for Document Management System

3.2.1 Use Case Scenarios

1. Failed User Login

- **Scenario:** User enters incorrect credentials.
- **Error:** Authentication fails.
- **Solution:** Validate credentials and display meaningful error messages.

2. Unauthorized Access Attempt

- **Scenario:** Officer tries to permanently delete a file.
- **Error:** Permission denied.
- **Solution:** Enforce RBAC rules and display access restriction alerts.

3. File Upload Failure

- **Scenario:** Officer uploads an unsupported file type.
- **Error:** Upload rejected.
- **Solution:** Validate file size and format with user feedback.

4. Role Assignment Error

- **Scenario:** Manager assigns invalid permissions to a role.
- **Error:** Role creation fails.
- **Solution:** Validate permission combinations and improve UI validation.

5. File Restore Failure

- **Scenario:** Manager attempts to restore a file from Trash.
- **Error:** File not found or conflict occurs.
- **Solution:** Verify file existence and handle conflicts gracefully.

6. Folder Deletion Issue

- **Scenario:** Officer deletes a folder containing files.
- **Error:** Operation blocked.
- **Solution:** Prompt confirmation and ensure safe deletion logic.

7. Profile Update Error

- **Scenario:** User updates profile with invalid data.
- **Error:** Update fails.
- **Solution:** Validate input fields and confirm successful updates.

3.2.2 Use Case Model

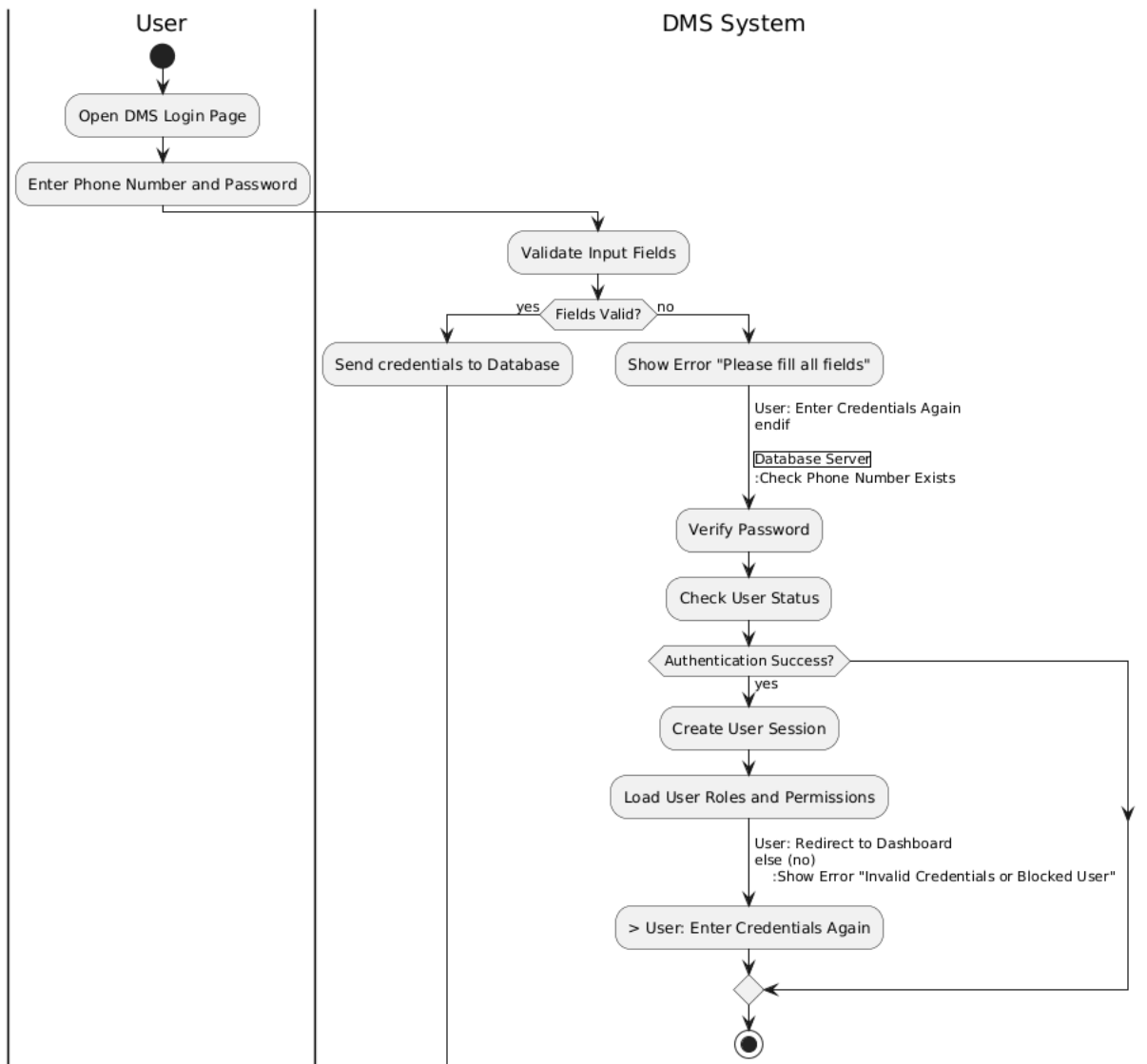
Primary Use Cases

1. User Login
2. User Logout
3. Manage Users
4. Manage Roles and Permissions
5. Create Folder
6. Edit Folder
7. Upload File
8. View File
9. Share File
10. Soft Delete File
11. Restore File
12. Permanently Delete File
13. View Trash
14. Profile Management

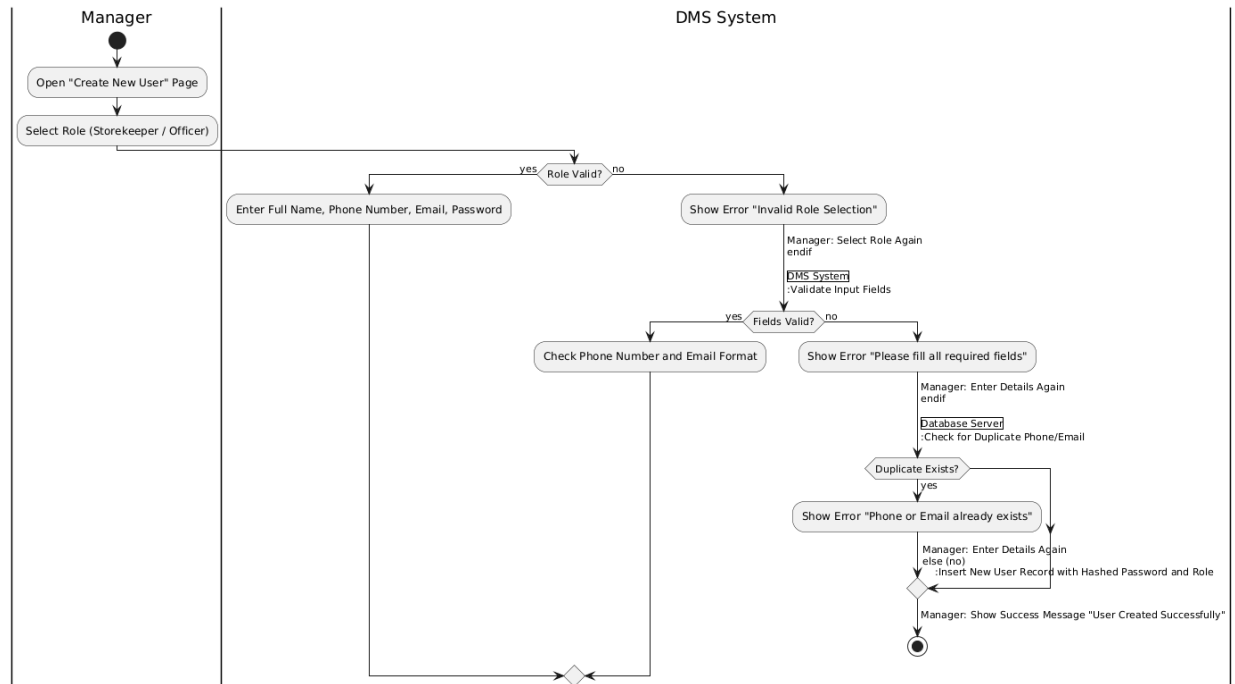
3.3 Activity Diagrams

Activity diagrams describe workflows and decision paths in the Document Management System.

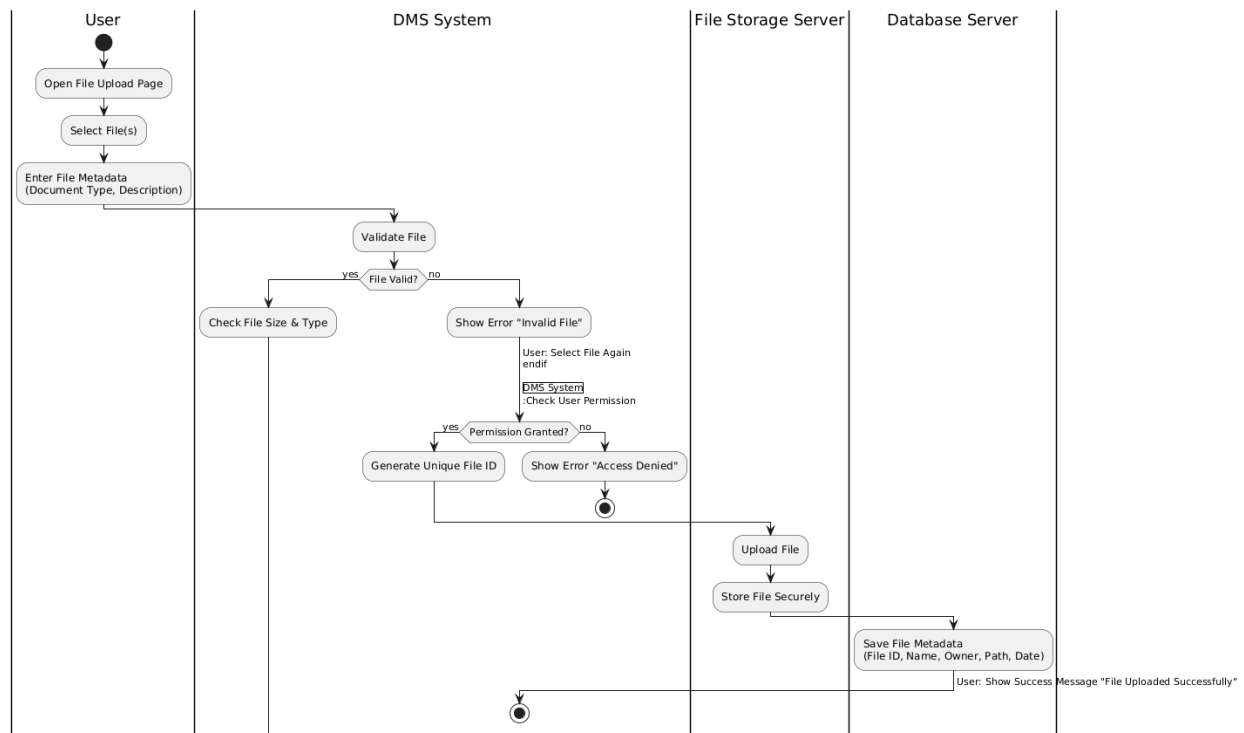
Key Activity Diagrams



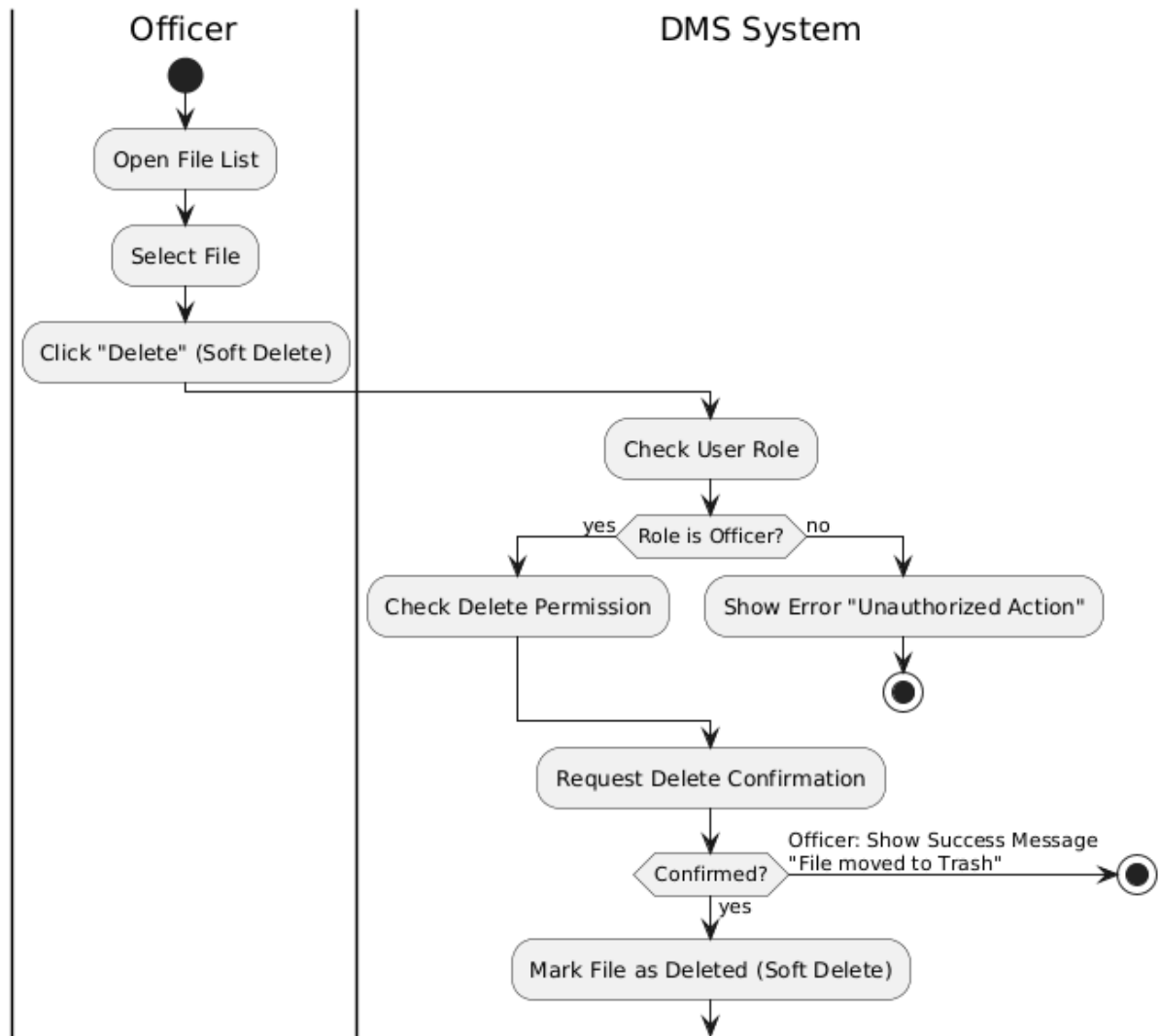
✦ **Figure 2:** Login Activity Diagram



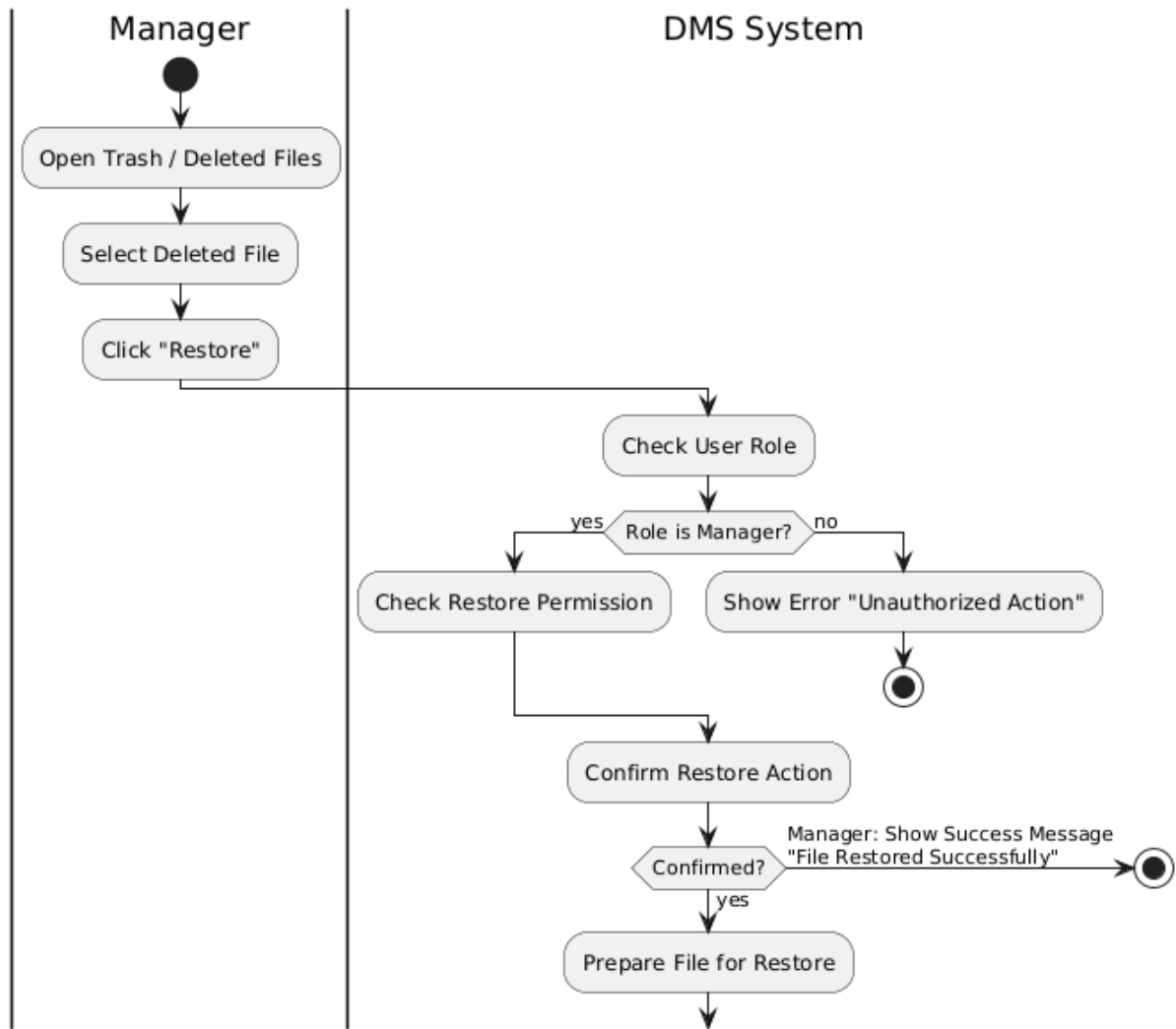
📌 **Figure 3: User Creation Activity Diagram**



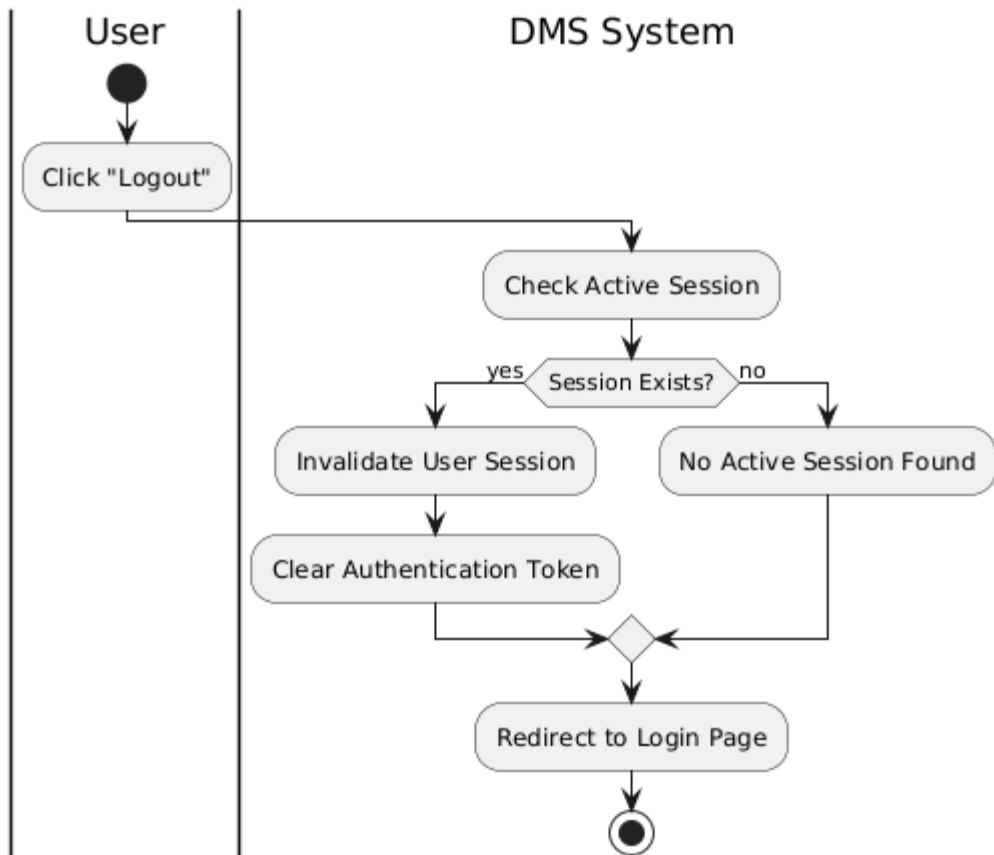
📌 **Figure 4: File Upload Activity Diagram**



✦ **Figure 5:** File Soft Delete Activity Diagram



✦ **Figure 6:** File Restore Activity Diagram



✦ **Figure 7: Logout Activity Diagram**

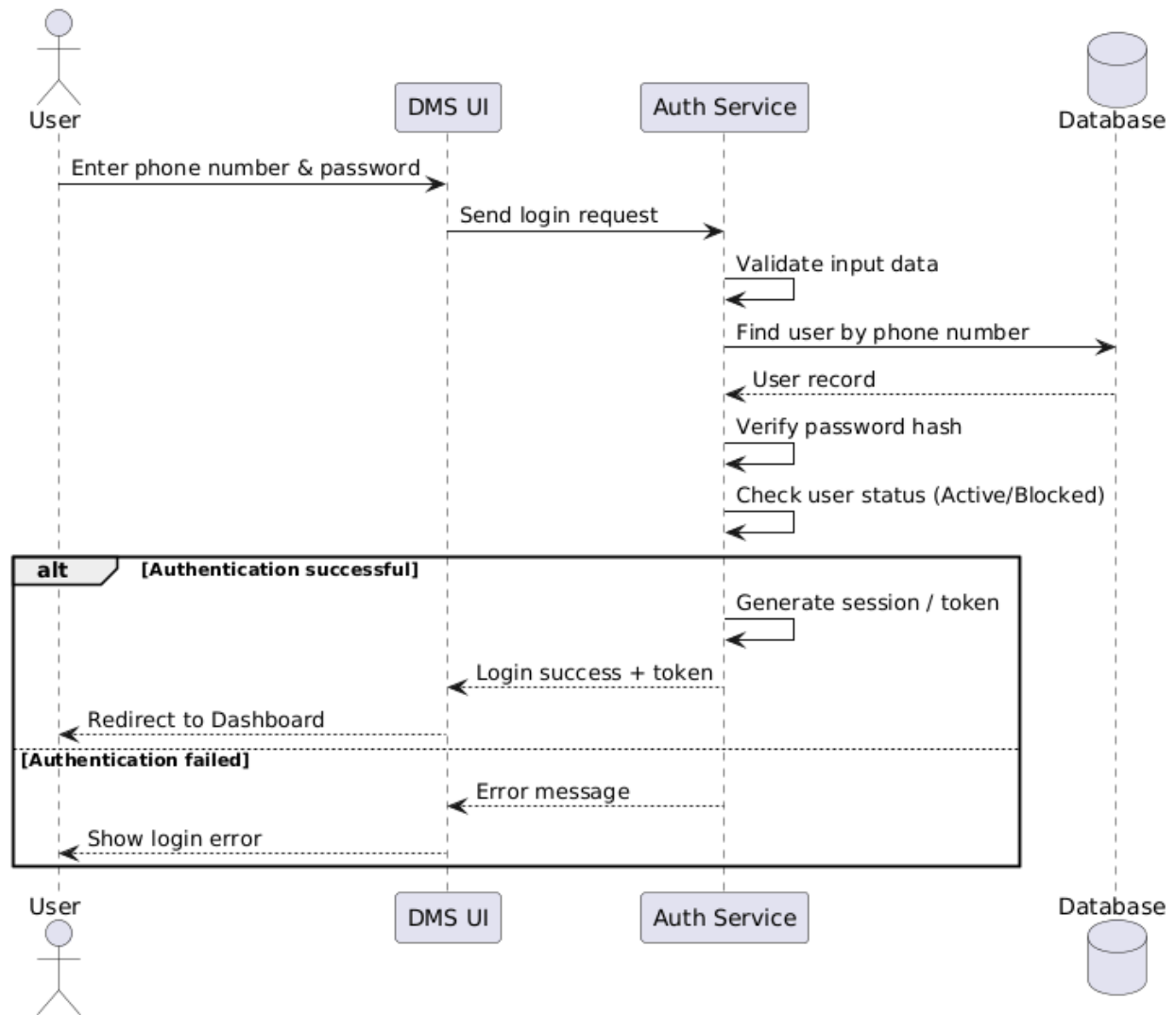
Each activity diagram demonstrates:

- Start and end points
- User decisions
- System validations
- Parallel or sequential actions

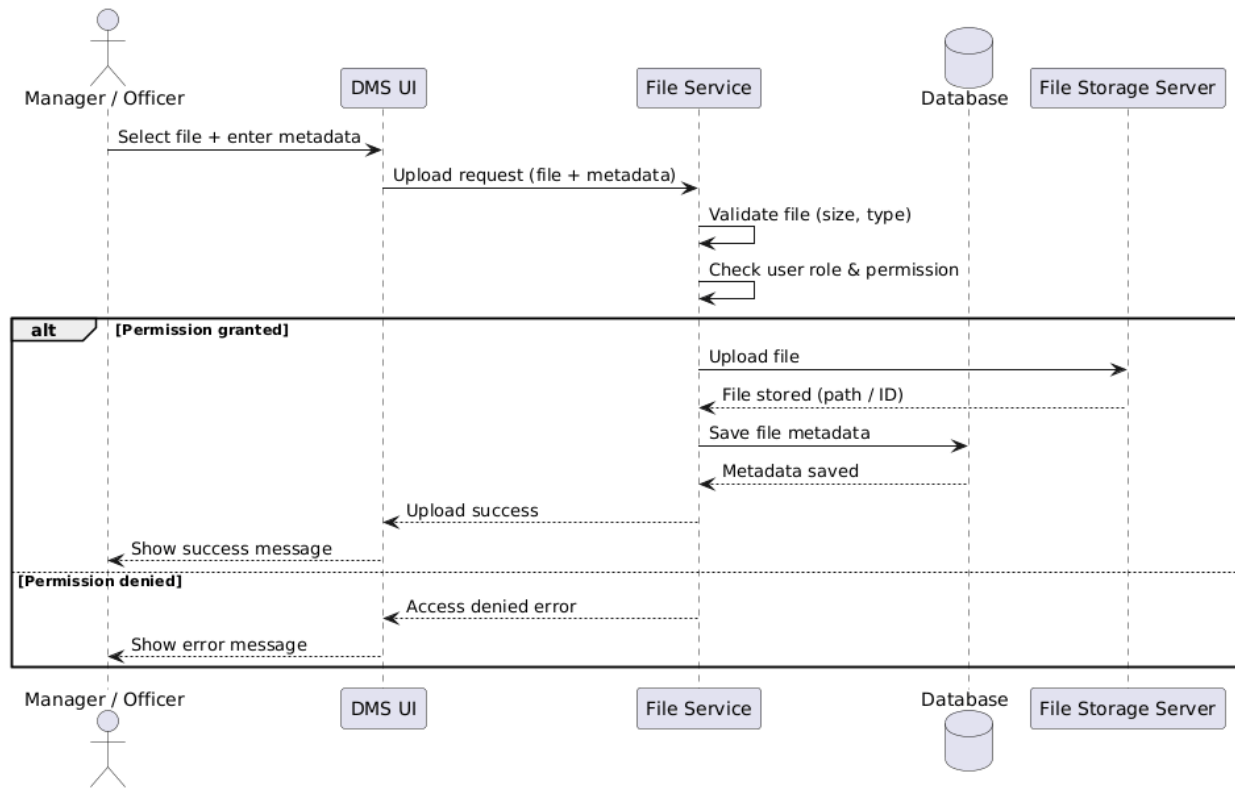
3.4 Sequence Diagrams

Sequence diagrams illustrate the interaction between users, frontend, backend, authentication service, and database during system operations.

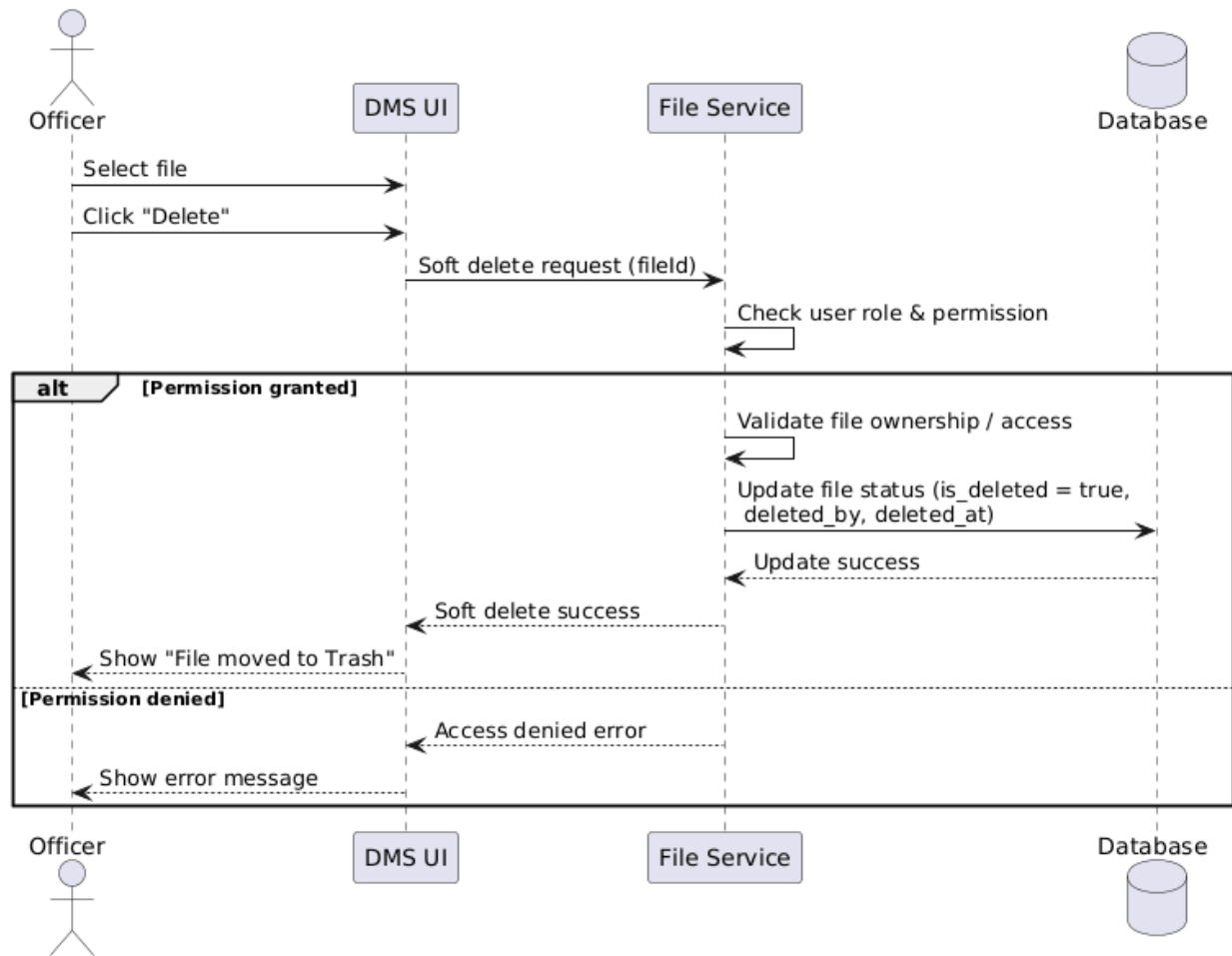
Key Sequence Diagrams



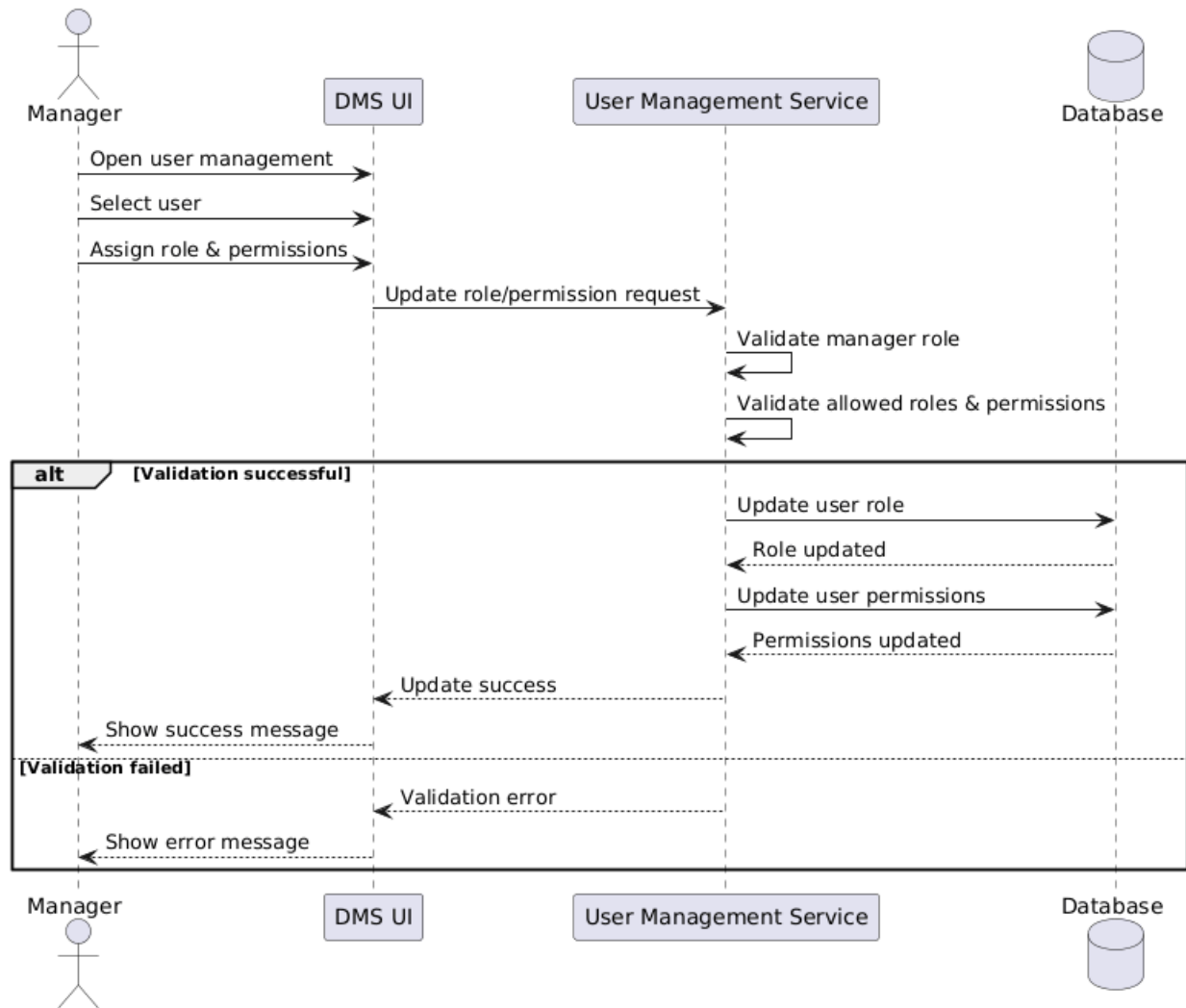
✦ **Figure 8:** Login Sequence Diagram



✦ **Figure 9:** File Upload Sequence Diagram



✦ **Figure 10:** File Soft Delete Sequence Diagram



✦ **Figure 11:** Role and Permission Assignment Sequence Diagram

CHAPTER 4

SYSTEM DESIGN (Document Management System)

4.1 Overview

The System Design phase of the **Document Management System (DMS)** defines how system requirements are translated into a secure, modular, and scalable architecture. The DMS is designed to manage digital files, folders, users, permissions, workflows, and audit trails while ensuring data integrity, security, and efficient document lifecycle management.

The design is guided by the need to support role-based access control, document versioning, approval workflows, and comprehensive auditing, as defined in the Prisma data model.

4.1.1 Design Considerations and Purposes

The primary purpose of the DMS design is to provide a centralized platform for managing documents securely across offices and users. The system supports multiple roles (Admin, Manager, Staff, User) and ensures controlled access to files using fine-grained permissions.

Key design considerations include:

- Secure authentication and role-based authorization
- Structured document storage using folders
- File versioning and workflow-based approvals
- File sharing with controlled access
- Audit logging for accountability and compliance

4.1.2 Design Goals

The main design goals of the Document Management System are:

- **User-Friendly Interface:**
Allow users to easily upload, organize, search, and manage files and folders.

- **Scalability:**
Support a growing number of users, files, versions, and workflows without performance degradation.
 - **Security:**
Protect sensitive documents using JWT-based authentication, role permissions, and file-level access control.
 - **Efficiency:**
Streamline document workflows such as submission, approval, version control, sharing, and archiving.
-

4.1.3 Design Guidance and Issues

4.1.3.1 Performance

The DMS is designed to handle concurrent users and large file volumes efficiently by:

- Indexing frequently accessed fields (userId, roleId, folderId)
 - Supporting asynchronous file uploads and processing
 - Optimizing database queries using Prisma ORM
 - Separating metadata storage from physical file storage
-

4.1.3.2 Dependability

System reliability is achieved through:

- Strong relational integrity using foreign keys
- File versioning to prevent data loss
- Soft delete functionality using the Trash module
- Comprehensive audit logging for every critical action

4.1.3.3 Maintenance

Maintainability is ensured through:

- Modular subsystem design
 - Clear separation of concerns
 - Well-defined Prisma schema and enums
 - Easy schema migration and extensibility
-

4.2 Proposed System Architecture

4.2.1 System Process

The operational flow of the Document Management System is as follows:

1. **User Authentication:**

Users authenticate using secure credentials and are assigned roles and permissions.

2. **Document Creation and Upload:**

Users upload files and organize them into folders. Each upload creates a new file version.

3. **Workflow Processing:**

Files move through workflow states such as Draft, Submitted, Approved, Rejected, or Archived.

4. **Permission Enforcement:**

File-level permissions (view, edit, delete, download) control access to documents.

5. **Sharing and Collaboration:**

Files can be shared securely using time-limited tokens.

6. **Audit and Monitoring:**

All actions are logged for transparency and accountability.

4.2.2 Subsystem Decomposition

The DMS is decomposed into the following subsystems:

- **User and Role Management:**

Handles users, roles, permissions, and access control.

- **Folder Management:**
Organizes files hierarchically using parent-child folder relationships.
- **File Management:**
Manages file metadata, ownership, and document attributes.
- **File Versioning:**
Stores multiple versions of a file with upload history.
- **Workflow Management:**
Controls document lifecycle states and approvals.
- **File Sharing:**
Enables secure file access using unique tokens.
- **Audit Logging:**
Tracks system activities for monitoring and compliance.
- **Trash Management:**
Supports soft deletion and document recovery.

4.2.3 Hardware/Software Mapping

Hardware:

- Application server
- Database server
- Client devices (desktop and mobile)
- Network infrastructure

Software:

- **Backend:** Node.js / Next.js with Prisma ORM
- **Frontend:** Next.js
- **Database:** MySQL
- **Authentication:** JWT with Role-Based Access Control (RBAC)

- **Deployment:** Apache Server (Ethio Telecom)
-

4.2.4 Persistent Data Management

Persistent data management is achieved through a relational database structure defined by the Prisma schema:

- Users, roles, and permissions stored relationally
 - Files and folders linked using foreign keys
 - File versions stored separately for traceability
 - Audit logs stored for accountability
 - Soft-deleted files preserved in Trash for recovery
-

4.5 Data Dictionary (Based on Prisma Schema)

User

- Stores user identity, authentication credentials, and role assignment.

Role & Permission

- Defines system access control using role-permission mapping.

Folder

- Represents hierarchical storage for files.

File

- Stores document metadata such as owner, folder, and personal information.

FileVersion

- Maintains document version history.

FilePermission

- Controls file-level access rights.

FileWorkflow

- Tracks document lifecycle states and approval actions.

FileShare

- Enables secure document sharing using tokens.

Tag & FileTag

- Supports document categorization and searching.

Trash

- Stores soft-deleted files for restoration.

AuditLog

- Records system activities for security and compliance.

This section describes the main data entities used in the **Document Management System (DMS)**. Each entity represents a database table defined in the Prisma schema and explains the purpose of key attributes.

4.5.1 User Entity

Name	Description	Data Type	Required
id	Unique identifier for user	String (UUID)	Yes
name	Full name of the user	String	Yes
phone	Phone number (unique)	String	Yes
username	Login username	String	Yes
password	Encrypted password	String	Yes
gender	User gender	Enum (MALE, FEMALE, OTHER)	Yes

Name	Description	Data Type	Required
status	User account status	Enum (ACTIVE, INACTIVE, PENDING, BLOCKED)	Yes
roleId	Reference to user role	String	Yes
createdAt	Account creation date	DateTime	Yes
updatedAt	Last update date	DateTime	Yes

4.5.2 Role Entity

Name	Description	Data Type	Required
id	Unique role identifier	String	Yes
name	Role name	String	Yes
description	Role description	String	No
createdAt	Role creation date	DateTime	Yes
updatedAt	Role update date	DateTime	Yes

4.5.3 Permission Entity

Name	Description	Data Type	Required
id	Permission identifier	String	Yes
code	Permission code	String	Yes
name	Permission name	String	Yes
description	Permission description	String	No
createdAt	Creation date	DateTime	Yes
updatedAt	Update date	DateTime	Yes

4.5.4 Folder Entity

Name	Description	Data Type Required	
id	Folder identifier	String	Yes
name	Folder name	String	Yes
description	Folder description	String	No
parentId	Parent folder ID	String	No
createdById	Folder creator	String	Yes
createdAt	Creation date	DateTime	Yes
updatedAt	Update date	DateTime	Yes

4.5.5 File Entity

Name	Description	Data Type Required	
id	File identifier	String	Yes
fileNumber	Unique file number	String	Yes
firstName	First name	String	Yes
fatherName	Father name	String	Yes
lastName	Last name	String	Yes
phone	Phone number	String	Yes
address	Address	String	Yes
folderId	Folder reference	String	No
createdById	File owner	String	Yes
createdAt	Creation date	DateTime	Yes
updatedAt	Update date	DateTime	Yes

4.5.6 FileVersion Entity

Name	Description	Data Type	Required
id	Version identifier	String	Yes
fileId	Reference to file	String	Yes
version	Version number	Integer	Yes
path	File storage path	String	Yes
size	File size	Integer	No
contentType	File type	String	No
uploadedById	Uploader ID	String	Yes
createdAt	Upload date	DateTime	Yes

4.5.7 FilePermission Entity

Name	Description	Data Type	Required
id	Permission ID	String	Yes
fileId	File reference	String	Yes
userId	User reference	String	Yes
permission	Access type	Enum (VIEW, EDIT, DELETE, DOWNLOAD)	Yes
grantedAt	Permission date	DateTime	Yes

4.5.8 FileWorkflow Entity

Name	Description	Data Type	Required
id	Workflow identifier	String	Yes
fileId	File reference	String	Yes
status	File status	Enum (DRAFT, SUBMITTED, APPROVED, REJECTED, ARCHIVED)	Yes
comment	Action note	String	No

Name	Description	Data Type	Required
actionById	Action performer	String	Yes
actionAt	Action time	DateTime	Yes

4.5.9 FileShare Entity

Name	Description	Data Type	Required
id	Share identifier	String	Yes
fileId	File reference	String	Yes
token	Unique access token	String	Yes
expiresAt	Expiry date	DateTime	No
isActive	Share status	Boolean	Yes
createdById	Share creator	String	Yes
createdAt	Creation date	DateTime	Yes

4.5.10 Trash Entity

Name	Description	Data Type	Required
id	Trash identifier	String	Yes
fileId	Deleted file reference	String	No
deletedBy	User who deleted file	String	Yes
deletedAt	Deletion date	DateTime	Yes
fileData	Stored file metadata	JSON	No

4.5.11 AuditLog Entity

Name	Description	Data Type	Required
id	Log identifier	Integer	Yes
timestamp	Action time	DateTime	Yes

Name	Description	Data Type Required	
actor	User name	String	Yes
role	User role	String	No
action	Performed action	String	Yes
resource	Affected resource	String	Yes
status	Action result	String	Yes
metadata	Additional data	JSON	No
userId	User reference	String	No

4.6 Cost Analysis

This section describes the estimated cost associated with the development, deployment, maintenance, and hosting of the **Document Management System (DMS)**. The cost model is designed to be transparent, affordable, and suitable for organizational use.

4.6.1 Software Development Cost

The software development cost covers the complete lifecycle of the system, starting from requirement analysis up to deployment.

This includes:

- System analysis and design
- Backend and frontend development
- Database design and implementation
- Authentication and authorization setup
- Testing and debugging
- Deployment and initial configuration

The development cost is considered a **it pay by divided the project payment in three phasee.**

Item	Description
Development Cost	1,500,000 ETB (One Million Five Hundred Thousand Birr)
Payment Type	Three phase payment(by divided into 3 phasee)
Coverage	Development → Testing → Deployment

4.6.2 Maintenance and Bug Fixing Cost

After deployment, the development team holds full responsibility for:

- Fixing system errors
- Debugging issues
- Resolving software bugs
- Ensuring system stability

This maintenance support applies **only to errors and bugs** found in the existing system features.

✓ **Bug fixing and error maintenance are included in the original payment until 1 years.**

✓ **No additional cost is required for fixing system defects**

However:

- If the organization requests **new features, additional functionalities, or system upgrades** beyond the original scope,
- These changes will be handled as **optional enhancements** and require **separate negotiation and agreement.**

4.6.3 Additional Feature Development Cost

Feature Type	Cost Policy
Bug fixing	Free (included)
System errors	Free (included)
Performance optimization	Free (included)

Feature Type	Cost Policy
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New features	Negotiable
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Major system expansion	Negotiable
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All additional features will be implemented **only after mutual agreement** between the client and the development team.

4.6.4 Server and Hosting Cost

The system will be hosted on a **VPS (Virtual Private Server)** provided by **Ethio Telecom**.

Server costs depend on:

- File types (documents, images, PDFs)
- File size and storage usage
- Number of system users
- System traffic and performance needs

Item	Description
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Server Provider	Ethio Telecom
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Server Type	VPS Server(4vcpu with 8 gb ram and must be a linux server)
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Storage	Scalable (expandable when needed)
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Cost Model	Periodic payment (monthly/yearly)
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The screenshot displays the Ethio Telecom Cloud Service console. At the top, there is a navigation bar with the Ethio Telecom logo, a search icon, and links for Contact Us, Console, Fuadekoo, and user icons. Below the navigation bar, a menu includes Home, Activities, Cloud Service, Solutions, Support, and Help. The main content area is a configuration form for a server. It includes sections for System Disk (set to ECS Volume (SAS) at 300 GB), Data Disk (set to ECS Volume (SAS) at 0 GB, with an option to add more), Quantity (set to 1), Duration (set to 1 month), Auto Renewal (set to Yes), and Currency (set to ETB). At the bottom, the final price is listed as 83010.84 ETB, with a note that the price is tax exclusive. Action buttons for Cancel, Buy Now, and Add To Cart are also present.

The server resources can be **upgraded at any time** based on system growth and storage requirements. The cost of the server from 83,010-200,000 ETB, this is a cost of the server from the ethio telecom

4.6.6 Overall Cost Evaluation

The overall cost of the Document Management System is **cost-effective** due to:

- Three phase development payment
- Free maintenance for system errors for defined time
- Scalable server infrastructure from Ethio Telecom

This cost model ensures long-term system sustainability while providing flexibility for future system expansion.

CHAPTER 5

CONCLUSION

In conclusion, the **Document Management System (DMS)** provides a comprehensive, secure, and scalable solution for managing digital documents within an organization. The system is

designed to handle the complete document lifecycle, from creation and upload to approval, sharing, archiving, and auditing.

The project successfully implemented advanced features such as role-based access control, file-level permissions, document versioning, workflow management, secure file sharing, and audit logging. These features ensure data integrity, accountability, and efficient collaboration among users.

Non-functional requirements, including security, performance, scalability, and maintainability, were prioritized throughout the system design. The modular architecture and well-defined Prisma schema allow for easy maintenance and future enhancements.

Through detailed modeling and system design, the complexity of the DMS was effectively captured and translated into a robust implementation. The system enhances productivity, reduces document loss, improves transparency, and provides a reliable foundation for future digital transformation initiatives.

Overall, the Document Management System stands as a modern and efficient platform that meets organizational needs for secure and structured document management.